



Experiment No.: 6

Title: Performance Testing



Aim: To perform performance testing using JMeter.

Resources needed: Internet, Apache JMeter

Theory:

The **Apache JMeter** is pure Java **open source** software, which was first developed by Stefano Mazzocchi of the Apache Software Foundation, designed to load test functional behaviour and measure performance. You can use JMeter to analyze and measure the performance of web applications or a variety of services. Performance Testing means testing a web application against heavy load, multiple and concurrent user traffic. JMeter originally is used for testing Web Application or FTP application. Nowadays, it is used for functional tests, database server tests etc.

Apache JMeter may be used to test performance both on static and dynamic resources, Web dynamic applications. It can be used to simulate a heavy load on a server, group of servers, network or object to test its strength or to analyze overall performance under different load types.

Procedure:

1. Install Java

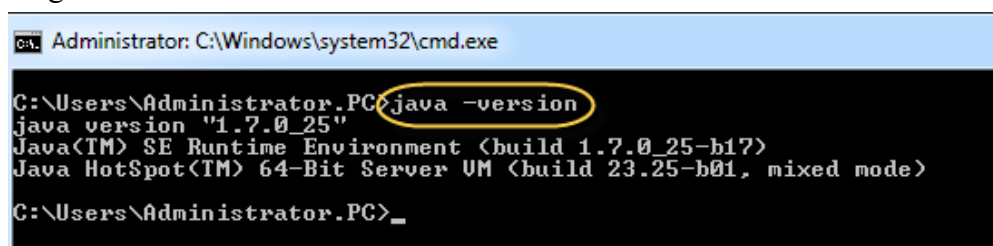
Because JMeter is a pure Java desktop application, it requires a fully compliant JVM 6 or higher. You can download and install the latest version of Java SE Development Kit.

Link: https://www.java.com/en/download/help/windows_manual_download.xml

After installation is finished, you can use the following procedure to check whether Java JDK is installed successfully in your system

- In Window/Linux, go to Terminal
- Enter command `java -version`

If Java runtime environment is installed successfully, you will see the output as figure below



```
Administrator: C:\Windows\system32\cmd.exe
C:\Users\Administrator.PC>java -version
java version "1.7.0_25"
Java(TM) SE Runtime Environment (build 1.7.0_25-b17)
Java HotSpot(TM) 64-Bit Server VM (build 23.25-b01, mixed mode)
C:\Users\Administrator.PC>
```

If nothing displays, please re-install Java SE runtime environment

2. Download JMeter

As of this writing, the latest version of JMeter is Apache JMeter 4.0.

Link: http://jmeter.apache.org/download_jmeter.cgi

Choose the Binaries file (either zip or tgz) to download as shown in figure below

Binaries

[apache-jmeter-4.0.tgz md5 sha512 pgp](#)
[apache-jmeter-4.0.zip md5 sha512 pgp](#)

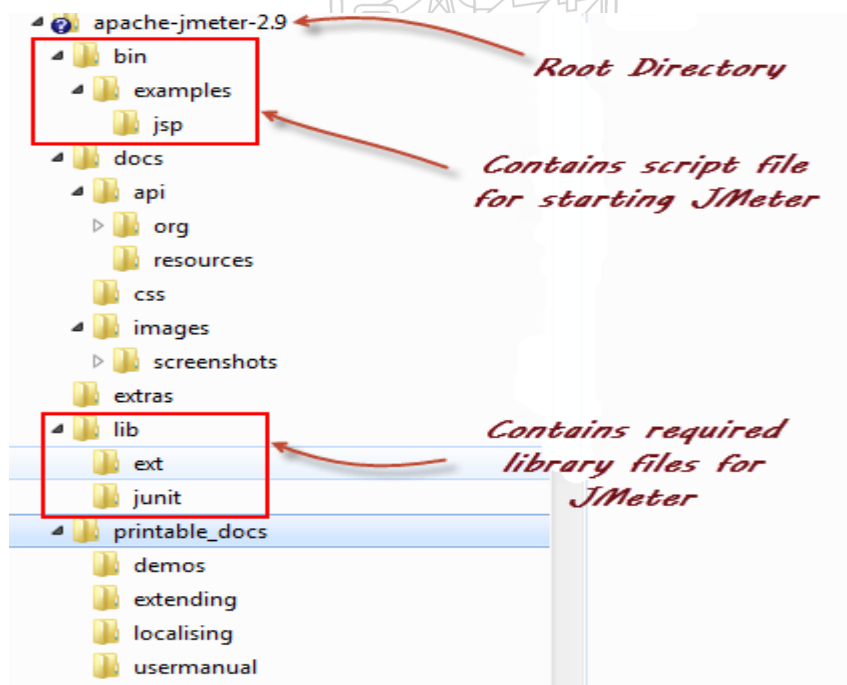
Source

[apache-jmeter-4.0_src.tgz md5 sha512 pgp](#)
[apache-jmeter-4.0_src.zip md5 sha512 pgp](#)

3. Installation

Installation of JMeter is extremely easy and simple. You simply unzip the zip/tar file into the directory where you want JMeter to be installed. There is no tedious installation screen to deal with. Simple unzip and you are done.

Once the unzipping is done installation directory structure should look like as figure below:



Given below is the description of the JMeter directories and its importance. JMeter directory contains many files and directory

- /bin: Contains JMeter script file for starting JMeter
- /docs: JMeter documentation files
- /extras: ant related extra files
- /lib/: Contains the required Java library for JMeter

- /lib/ext: contains the core jar files for JMeter and the protocols
- /lib/junit: [JUnit](#) library used for JMeter
- /printable_docs:

4. Launch JMeter

You can start JMeter in two modes:

- GUI Mode
- Server Mode

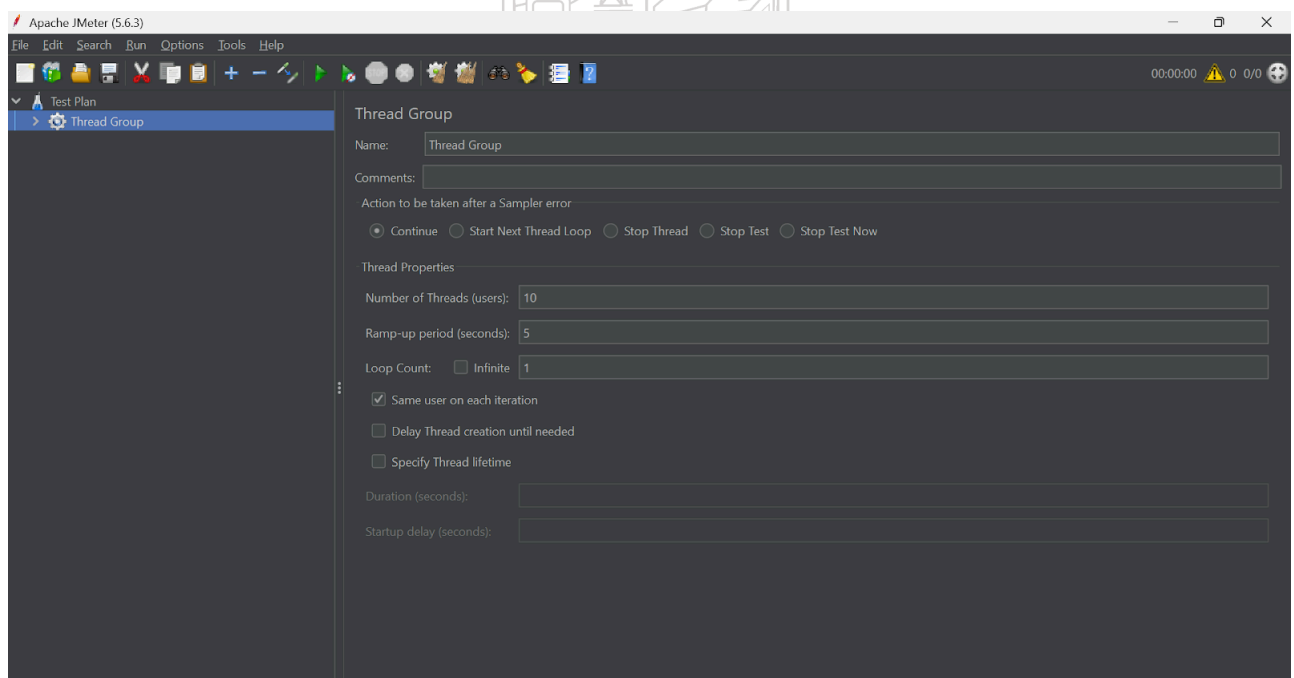
Start JMeter in GUI Mode

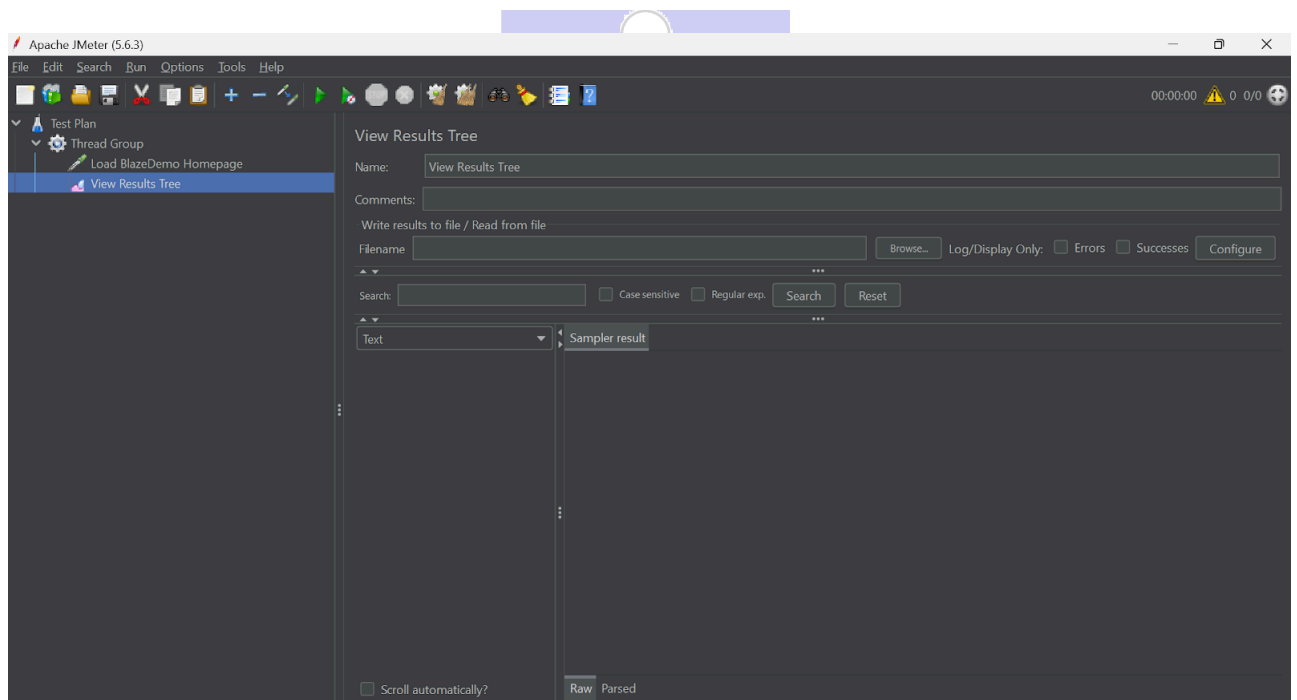
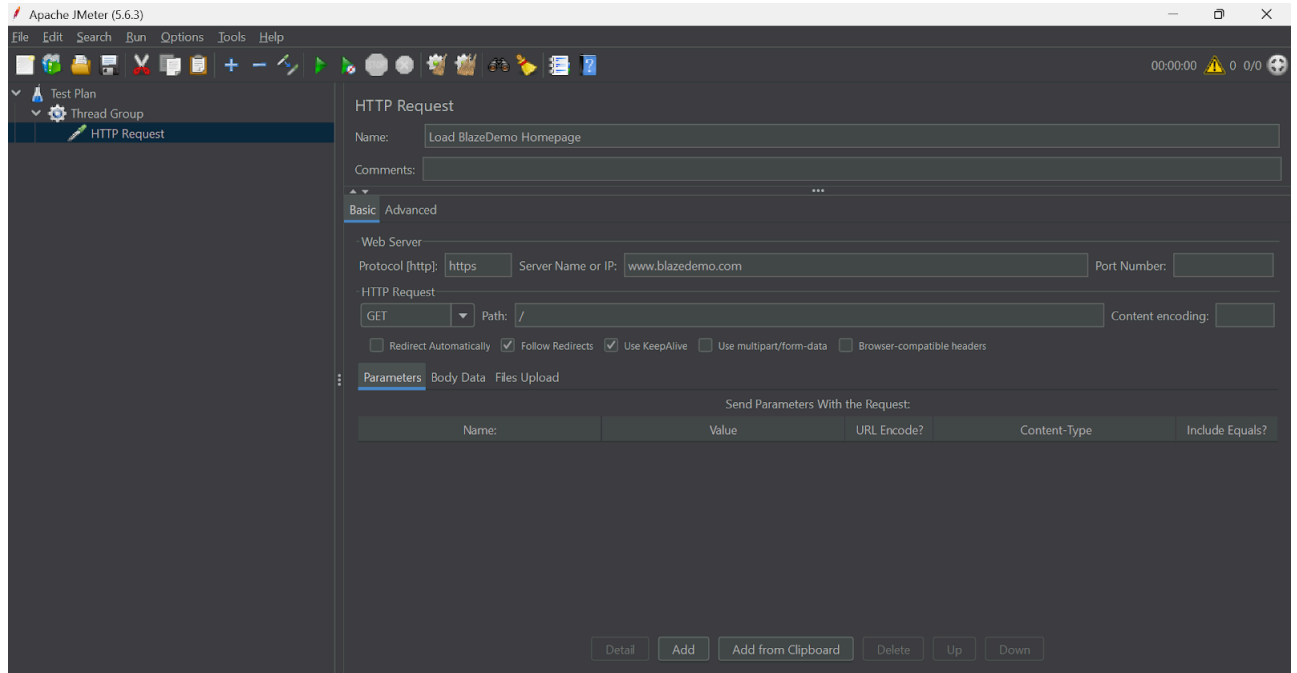
If you are using Window, just run the file /bin/jmeter.bat to start JMeter in GUI mode.

Perform the following on Apache JMeter:

1. Create a Performance Test Plan
2. Use Constant Timer
3. Use Response Assertion
4. Loop Controller
5. Post Processor

Results: (Document printout as per the format)





Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

Test Plan

Thread Group

Load BlazeDemo Homepage

View Results Tree

Summary Report

Summary Report

Name: Summary Report

Comments:

Write results to file / Read from file

Filename

Browse...

Log/Display Only: ☐ Errors ☐ Successes

Configure

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB...	Sent KB/sec	Avg. Bytes
TOTAL	0	0	#N/A	#N/A	0.00	0.00%	.0/hour	0.00	0.00	.0

☐ Include group name in label?

Save Table Data

☒ Save Table Header

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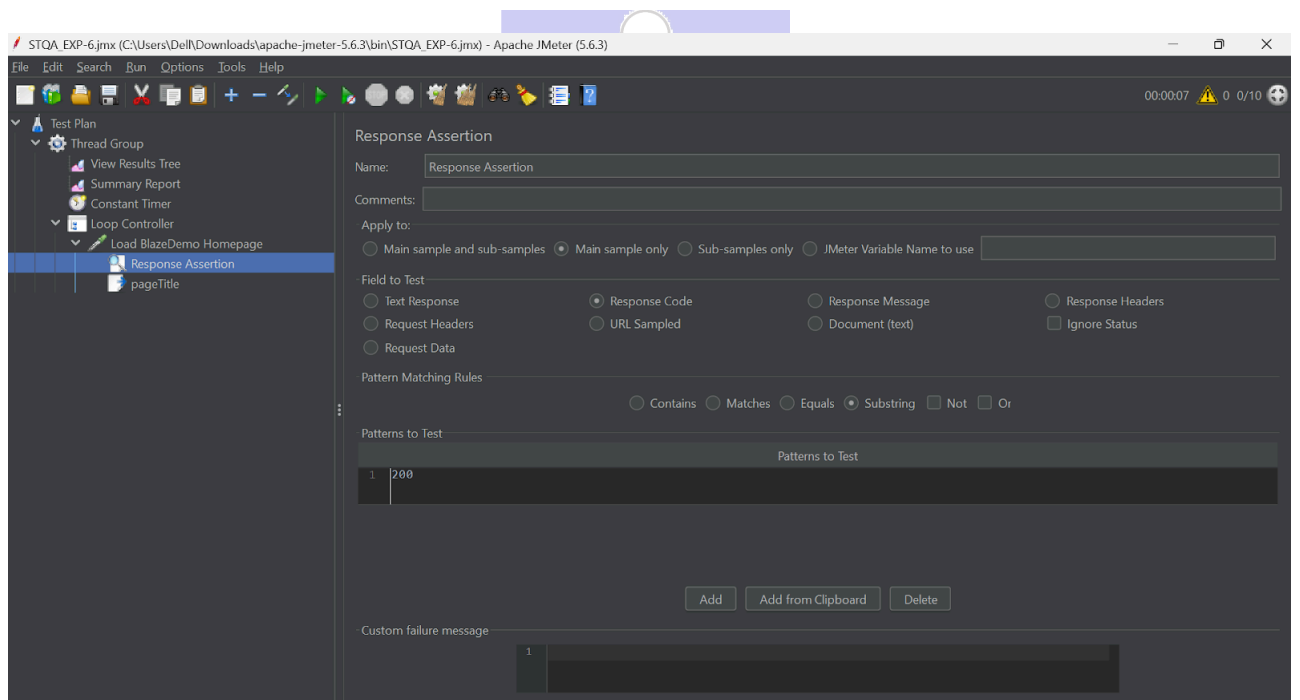
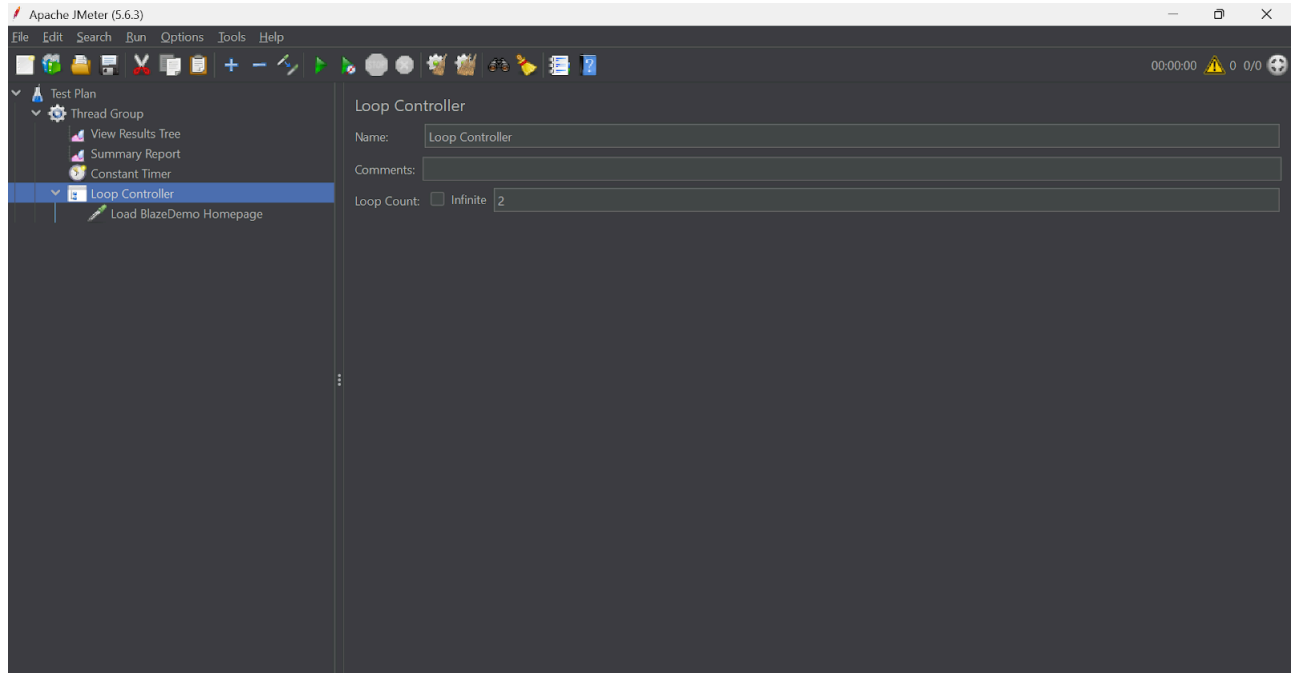
Constant Timer

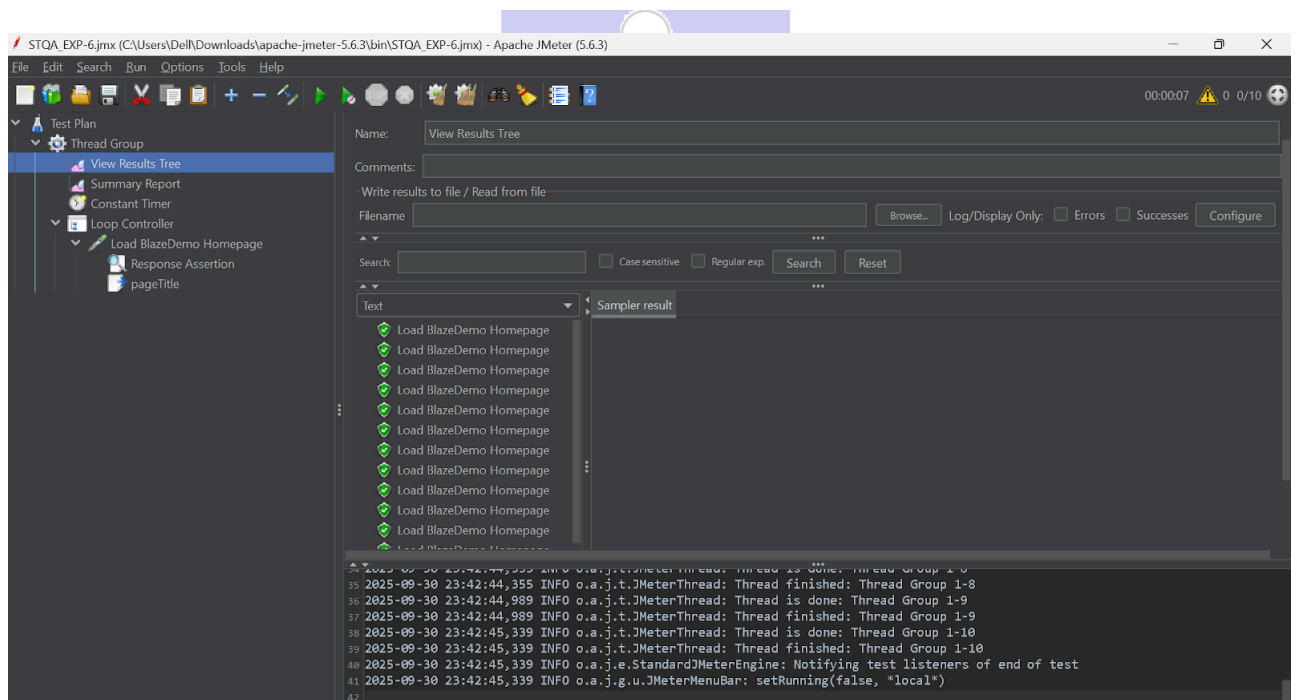
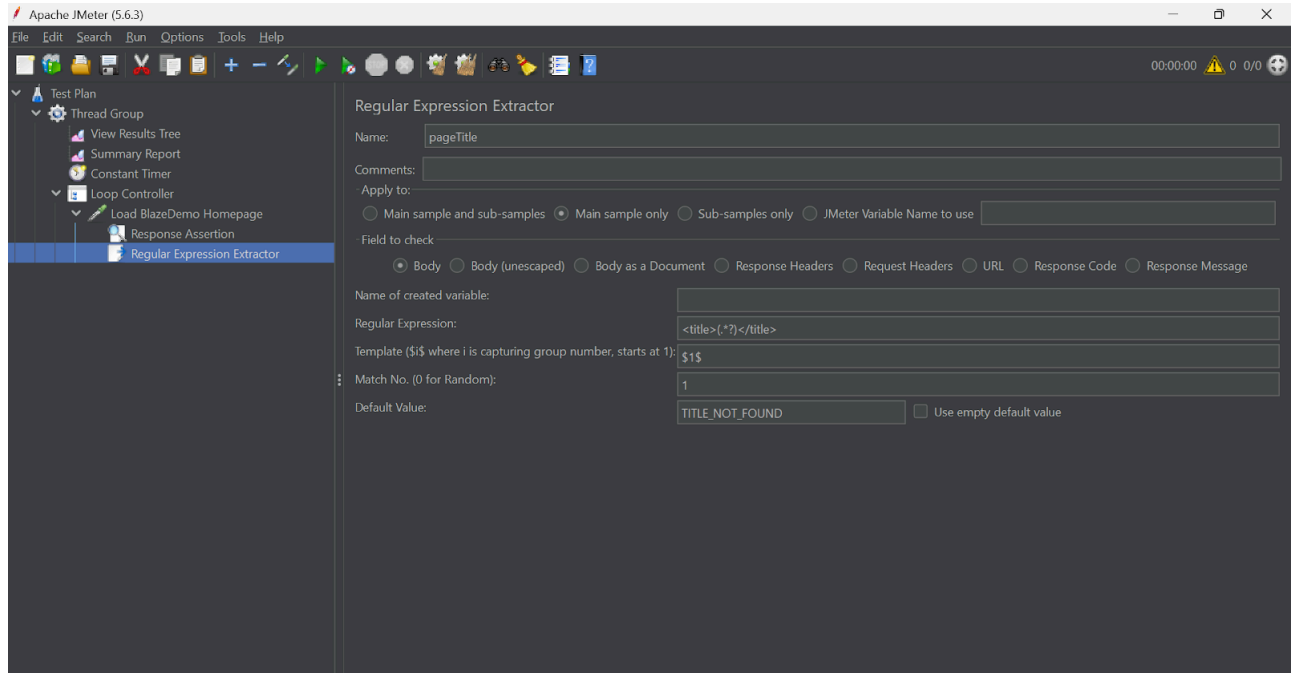
Constant Timer

Name: Constant Timer

Comments:

Thread Delay (in milliseconds): 1000





Summary Report

Name: Summary Report

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
Load BlazeD...	20	444	361	566	59.20	0.00%	3.2/sec	14.73	0.37	4757.0
TOTAL	20	444	361	566	59.20	0.00%	3.2/sec	14.73	0.37	4757.0

☐ Include group name in label? ☒ Save Table Header

Questions:

1. Why is performance testing considered highly important for web applications? Outline the performance parameters against which testing can be performed.

Ans: Performance testing is crucial for web applications because it directly impacts user satisfaction, brand reputation, and revenue. A slow or unreliable application leads to a poor user experience, causing users to abandon the site, which can result in lost sales and a damaged reputation. It ensures the application is:

- **Speedy and Responsive:** Users expect fast load times.
- **Scalable:** The application can handle a growing number of users or sudden traffic spikes without crashing.
- **Stable:** The application is reliable under the expected user load.

Key performance parameters include:

- **Response Time:** The total time from a user sending a request until they receive the full response.
- **Throughput:** The number of requests the application can handle per unit of time (e.g., requests per second).
- **Latency:** The time it takes for the first byte of the response to be delivered after a request is sent.
- **Error Rate:** The percentage of requests that result in an error compared to the total number of requests.
- **CPU Utilization:** The percentage of CPU time used to process requests.
- **Memory Utilization:** The amount of memory used by the application while under load.

2. What are the major challenges in performance testing?

Ans: The major challenges in performance testing include:

- **Realistic Test Environment:** Creating a test environment that perfectly mimics the live production environment is often complex and expensive.
- **Simulating User Behavior:** Accurately simulating how real users interact with the application, including their navigation paths and usage patterns, can be difficult.
- **Identifying Bottlenecks:** Pinpointing the exact root cause of a performance issue (e.g., database query, network latency, inefficient code) can be a significant challenge.
- **Tool Expertise:** Performance testing tools like JMeter are powerful but require specific knowledge and skills to be used effectively.
- **Managing Test Data:** Generating and maintaining large volumes of realistic and secure data for testing can be complicated.

Outcomes: CO3 – Apply recent automation tools for testing software.

Conclusion: (Conclusion to be based on outcomes)

This experiment successfully demonstrated the use of Apache JMeter to conduct performance testing on a web application. By creating a test plan, adding thread groups to simulate user load, and using elements like timers and assertions, we were able to measure and analyze key performance metrics. This practical application of JMeter fulfills the objective of applying recent automation tools for software testing, providing valuable insights into application behavior under stress and ensuring a better end-user experience.

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of faculty in-charge with date

Reference Websites:

1. <https://www.guru99.com/jmeter-performance-testing.html>
 2. <https://dzone.com/articles/jmeter-performance-and-load-testing>
 3. <https://jmeter.apache.org/>
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